

# Appendix

## Semantic Tableaux

| $\alpha$                    | $\alpha_1$            | $\alpha_2$            | $\beta$                         | $\beta_1$                   | $\beta_2$                   |
|-----------------------------|-----------------------|-----------------------|---------------------------------|-----------------------------|-----------------------------|
| $\neg\neg A_1$              | $A_1$                 |                       | $\neg(B_1 \wedge B_2)$          | $\neg B_1$                  | $\neg B_2$                  |
| $A_1 \wedge A_2$            | $A_1$                 | $A_2$                 | $B_1 \vee B_2$                  | $B_1$                       | $B_2$                       |
| $\neg(A_1 \vee A_2)$        | $\neg A_1$            | $\neg A_2$            | $B_1 \rightarrow B_2$           | $\neg B_1$                  | $B_2$                       |
| $\neg(A_1 \rightarrow A_2)$ | $A_1$                 | $\neg A_2$            | $B_1 \uparrow B_2$              | $\neg B_1$                  | $\neg B_2$                  |
| $\neg(A_1 \uparrow A_2)$    | $A_1$                 | $A_2$                 | $\neg(B_1 \downarrow B_2)$      | $B_1$                       | $B_2$                       |
| $A_1 \downarrow A_2$        | $\neg A_1$            | $\neg A_2$            | $\neg(B_1 \leftrightarrow B_2)$ | $\neg(B_1 \rightarrow B_2)$ | $\neg(B_2 \rightarrow B_1)$ |
| $A_1 \leftrightarrow A_2$   | $A_1 \rightarrow A_2$ | $A_2 \rightarrow A_1$ | $B_1 \oplus B_2$                | $\neg(B_1 \rightarrow B_2)$ | $\neg(B_2 \rightarrow B_1)$ |
| $\neg(A_1 \oplus A_2)$      | $A_1 \rightarrow A_2$ | $A_2 \rightarrow A_1$ |                                 |                             |                             |

  

| $\gamma$             | $\gamma(a)$ |
|----------------------|-------------|
| $\forall x.A(x)$     | $A(a)$      |
| $\neg\exists x.A(x)$ | $\neg A(a)$ |

  

| $\delta$             | $\delta(a)$ |
|----------------------|-------------|
| $\exists x.A(x)$     | $A(a)$      |
| $\neg\forall x.A(x)$ | $\neg A(a)$ |

## Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$
  

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$  is a *sequent* and means that if  $A$  is true, then  $B$  can be inferred.

| $\alpha$                        | $\alpha_1$                  | $\alpha_2$                  | $\beta$                     | $\beta_1$             | $\beta_2$             |
|---------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------|-----------------------|
| $\neg\neg A$                    | $A$                         |                             | $B_1 \wedge B_2$            | $B_1$                 | $B_2$                 |
| $\neg(A_1 \wedge A_2)$          | $\neg A_1$                  | $\neg A_2$                  | $\neg(B_1 \vee B_2)$        | $\neg B_1$            | $\neg B_2$            |
| $A_1 \vee A_2$                  | $A_1$                       | $A_2$                       | $\neg(B_1 \rightarrow B_2)$ | $B_1$                 | $\neg B_2$            |
| $A_1 \rightarrow A_2$           | $\neg A_1$                  | $A_2$                       | $\neg(B_1 \uparrow B_2)$    | $B_1$                 | $B_2$                 |
| $A_1 \uparrow A_2$              | $\neg A_1$                  | $\neg A_2$                  | $\neg(B_1 \downarrow B_2)$  | $\neg B_1$            | $\neg B_2$            |
| $A_1 \downarrow A_2$            | $A_1$                       | $A_2$                       | $B_1 \leftrightarrow B_2$   | $B_1 \rightarrow B_2$ | $B_2 \rightarrow B_1$ |
| $\neg(A_1 \leftrightarrow A_2)$ | $\neg(A_1 \rightarrow A_2)$ | $\neg(A_2 \rightarrow A_1)$ | $\neg(B_1 \oplus B_2)$      | $B_1 \rightarrow B_2$ | $B_2 \rightarrow B_1$ |
| $A_1 \oplus A_2$                | $\neg(A_1 \rightarrow A_2)$ | $\neg(A_2 \rightarrow A_1)$ |                             |                       |                       |

  

| $\gamma$             | $\gamma(a)$ |
|----------------------|-------------|
| $\exists x.A(x)$     | $A(a)$      |
| $\neg\forall x.A(x)$ | $\neg A(a)$ |

  

| $\delta$             | $\delta(a)$ |
|----------------------|-------------|
| $\forall x.A(x)$     | $A(a)$      |
| $\neg\exists x.A(x)$ | $\neg A(a)$ |